CHAPTER 13 · DERIVATIONS

ELECTROSTATICS

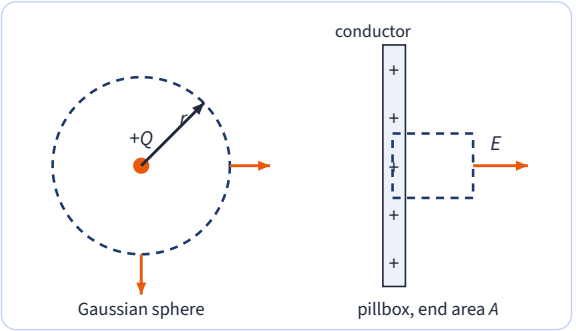
2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 ELECTRIC FIELD STRENGTH FROM GAUSS'S LAW

SYLLABUS 13.3(D)

RESULT $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; sphere: 0 inside; plate: $E = \dfrac{\sigma}{\varepsilon_0}$

1. **Point charge Q**: Gaussian sphere of radius r centred on Q. E is radial and the same everywhere on it, so $\Phi = E(4\pi r^2) = \dfrac{Q}{\varepsilon_0}$, giving $\mathbf{E = \dfrac{Q}{4\pi\varepsilon_0 r^2}}$.
2. **Charged conducting sphere** (radius R, charge Q on its surface): for r > R the Gaussian sphere encloses Q, so $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; for r < R it encloses no charge, so **E = 0 inside**.
3. **Charged conducting plate** (surface charge density σ): a pillbox of end area A crosses the surface, one end inside the conductor (E = 0) and one end outside.
4. Flux only through the outer end: $EA = \dfrac{\sigma A}{\varepsilon_0}$, so $\mathbf{E = \dfrac{\sigma}{\varepsilon_0}}$, perpendicular to the surface.



2 GAUSS'S LAW IS EQUIVALENT TO COULOMB'S LAW

SYLLABUS 13.3

RESULT $F = Qq/(4\pi\varepsilon_0 r^2)$

1. Draw a Gaussian surface: a sphere of radius r centred on a point charge Q.
2. Apply Gauss's law: total flux through the closed surface $\Phi = \dfrac{Q}{\varepsilon_0}$.
3. Definition of electric flux: E is radial and the same at every point of the sphere, so $\Phi = E(4\pi r^2)$.
4. $E(4\pi r^2) = \dfrac{Q}{\varepsilon_0}$, so the electric field strength is $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$.
5. Place a charge q at distance r and use the definition $E = \dfrac{F}{q}$: $F = qE$, so $\mathbf{F = Qq/(4\pi\varepsilon_0 r^2)}$, which is Coulomb's law.

CHAPTER 14 · DERIVATIONS

CAPACITORS

5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CAPACITANCE OF A PARALLEL PLATE CAPACITOR

SYLLABUS 14.1(D)

RESULT $C_0 = \dfrac{\varepsilon_0 A}{d}$

1. Plates of area A, separation d, charges ±Q: $\sigma = \dfrac{Q}{A}$.
2. Gauss's law: the field between the plates is $E = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 A}$ (uniform).
3. $V = Ed = Qd/(\varepsilon_0 A)$.
4. $\mathbf{C_0 = \dfrac{Q}{V} = \dfrac{\varepsilon_0 A}{d}}$.

2 CAPACITORS IN SERIES AND IN PARALLEL

SYLLABUS 14.3(A)

RESULT series: $\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \ldots$; parallel: $C = C_1 + C_2 + \ldots$

1. **Series**: each capacitor carries the same charge Q, and $V = V_1 + V_2 + V_3$.
2. $\dfrac{Q}{C} = \dfrac{Q}{C_1} + \dfrac{Q}{C_2} + \dfrac{Q}{C_3}$, so $\mathbf{\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3}}$.
3. **Parallel**: each capacitor has the same p.d. V, and $Q = Q_1 + Q_2 + Q_3$.
4. $CV = C_1V + C_2V + C_3V$, so $\mathbf{C = C_1 + C_2 + C_3}$.

Energy formulae $U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \dfrac{Q^2}{2C}$: derivation not required (14.4).

3 WHY A DIELECTRIC INCREASES THE CAPACITANCE (QUALITATIVE)

SYLLABUS 14.3

RESULT $C = \dfrac{Q}{V}$ increases

1. When the dielectric is inserted, its positive nuclei are attracted towards the negative plate and its electrons towards the positive plate.
2. Dipoles are formed (the dielectric is polarised); the induced charges set up an electric field opposite to the applied field.
3. The net field between the plates decreases: the positive plate becomes less positive and the negative plate less negative in potential.
4. The potential difference V between the plates decreases while the charge Q remains constant (isolated capacitor).
5. From $C = \dfrac{Q}{V}$, the **capacitance increases**.

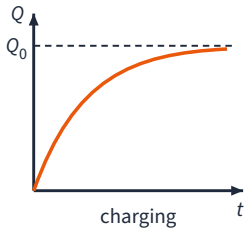
Points must be given in this order. If the capacitor stays connected to the battery, V is constant and Q increases instead.

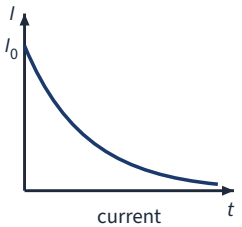
4 CHARGING THROUGH A RESISTOR

SYLLABUS 14.5(B)

RESULT $Q = Q_0(1 - e^{-t/RC})$, $V = V_0(1 - e^{-t/RC})$, $I = I_0e^{-t/RC}$

1. Kirchhoff's law round the circuit (supply V_0): $V_0 = IR + \frac{Q}{C}$, with $I = \frac{dQ}{dt}$.
2. $R\frac{dQ}{dt} = \frac{CV_0 - Q}{C}$, so $\frac{dQ}{CV_0 - Q} = \frac{dt}{RC}$.
3. Integrate from $Q = 0$ at $t = 0$: $-\ln\frac{CV_0 - Q}{CV_0} = \frac{t}{RC}$.
4. $\mathbf{Q = Q_0(1 - e^{-t/RC})}$, with $Q_0 = CV_0$; dividing by C: $\mathbf{V = V_0(1 - e^{-t/RC})}$.
5. $I = \frac{dQ}{dt} = \frac{Q_0}{RC}e^{-t/RC}$: $\mathbf{I = I_0e^{-t/RC}}$, with $I_0 = \frac{V_0}{R}$.





5 DISCHARGING THROUGH A RESISTOR

SYLLABUS 14.5(D)

RESULT $Q = Q_0e^{-t/RC}$, $V = V_0e^{-t/RC}$, $I = I_0e^{-t/RC}$

1. With the supply removed: $0 = IR + \frac{Q}{C}$, and $I = \frac{dQ}{dt}$.
2. $\frac{dQ}{dt} = -\frac{Q}{RC}$, so $\frac{dQ}{Q} = -\frac{dt}{RC}$.
3. Integrate from Q_0 at $t = 0$: $\ln\frac{Q}{Q_0} = -\frac{t}{RC}$, so $\mathbf{Q = Q_0e^{-t/RC}}$ and $\mathbf{V = V_0e^{-t/RC}}$.
4. Size of the current: $I = Q_0e^{-t/RC}/RC$, i.e. $\mathbf{I = I_0e^{-t/RC}}$ (in the opposite direction to the charging current).

CHAPTER 15 · DERIVATIONS

ELECTRIC CURRENT

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CURRENT AND DRIFT VELOCITY

SYLLABUS 15.2(B)

RESULT $I = Anev$

1. Wire of cross-sectional area A, n free electrons per unit volume, drift velocity v.
2. In time t the electrons move a distance vt, so all electrons in the volume Avt pass a cross-section.
3. Number = nAvt; charge Q = nAvte.
4. $\mathbf{I = \frac{Q}{t} = Anev}$.

2 OHM'S LAW AND J = ΣE

SYLLABUS 15.4(E)

RESULT $J = \sigma E \Leftrightarrow V = IR$

1. Uniform conductor of length l and area A with p.d. V: $J = \frac{I}{A}$ and $E = \frac{V}{l}$.
2. $J = \sigma E$ gives $\frac{I}{A} = \frac{\sigma V}{l}$.
3. $V = I\frac{l}{\sigma A} = I\frac{\rho l}{A} = IR$.
4. For a given conductor at constant temperature R is constant, so $V \propto I$: Ohm's law.

3 CONDUCTIVITY

SYLLABUS 15.4(F)

RESULT $\sigma = \frac{ne^2t}{m}$

1. In a field E an electron has acceleration $a = \frac{eE}{m}$.
2. Starting from zero average velocity after each collision, the mean time between collisions is t, so the drift velocity $v = at = \frac{eEt}{m}$.
3. $J = nev = \frac{ne^2t}{m}E$.
4. Comparing with $J = \sigma E$: $\mathbf{\sigma = \frac{ne^2t}{m}}$.

Use $\sigma = \frac{ne^2t}{m}$ to explain: a conductor conducts better than a semiconductor (much larger n); for a conductor, a higher temperature decreases t, so σ decreases; for a semiconductor, a higher temperature increases n, so σ increases.

4 RESISTORS IN SERIES AND IN PARALLEL

SYLLABUS 15.4

RESULT series: $R = R_1 + R_2 + \ldots$; parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

1. **Series**: the same current I flows through each resistor, but the potential differences are different.
2. Conservation of energy: $V = V_1 + V_2 + V_3$, so $IR = IR_1 + IR_2 + IR_3$ and $\mathbf{R = R_1 + R_2 + R_3}$.
3. **Parallel**: the potential difference V is the same across each resistor, but the currents are different.
4. Conservation of charge: $I = I_1 + I_2 + I_3$, so $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$ and $\mathbf{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$.

CHAPTER 16 · DERIVATIONS

DIRECT CURRENT CIRCUITS

2
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CONVERTING A GALVANOMETER: SHUNT AND MULTIPLIER

SYLLABUS 16.4

RESULT $R_s = I_g r_g/(I - I_g)$; $R_m = \dfrac{V}{I_g} - r_g$

1. A galvanometer (milliammeter) of resistance r_g gives full-scale deflection at current I_g.
2. **Ammeter** (range I): connect a shunt R_s of very low resistance in parallel. The p.d. is the same: $V_{shunt} = V_{meter}$, so $(I - I_g)R_s = I_g r_g$ and $\mathbf{R_s = I_g r_g/(I - I_g)}$.
3. **Voltmeter** (range V): connect a multiplier R_m of very high resistance in series. The current is the same: $V = V_{meter} + V_{multiplier} = I_g(r_g + R_m)$, so $\mathbf{R_m = \dfrac{V}{I_g} - r_g}$.

An ammeter must have a very low resistance and a voltmeter a very high resistance, so that they hardly change the circuit.

2 POTENTIOMETER CALCULATIONS

SYLLABUS 16.4

RESULT $V \propto l$

1. **Driver cell only** across the wire (no other resistance): the p.d. across a length l is proportional to l, so $\dfrac{E_1}{E_2} = \dfrac{l_1}{l_2}$.
2. **Driver cell with a resistance** (internal resistance r or a series resistor R): first find the current in the upper (driver) circuit, $I = \dfrac{E}{R_{wire} + R + r}$.
3. Then the p.d. per unit length of wire is $k = \dfrac{IR_{wire}}{L}$, and at balance the e.m.f. of the cell in the lower circuit is $E_x = k\,l$.

At balance no current flows through the galvanometer, so the potentiometer measures the e.m.f., not the terminal p.d.

CHAPTER 17 · DERIVATIONS

MAGNETIC FIELDS

7 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CHARGE MOVING ACROSS A UNIFORM ELECTRIC FIELD

SYLLABUS 17.2

RESULT $y = \frac{qEx^2}{2mu^2}$

1. A charge q enters a uniform field E with speed u perpendicular to the field.
2. Along the original direction there is no force: $x = ut$.
3. Along the field the force $qE = ma$ gives a constant acceleration $a = \frac{qE}{m}$: $y = \frac{1}{2}at^2$.
4. Eliminate t: $\mathbf{y = \frac{qEx^2}{2mu^2}}$, a parabolic path (unlike the circular path in a magnetic field, where $F \perp v$ and the speed is constant).

2 FORCE ON A CURRENT-CARRYING CONDUCTOR

SYLLABUS 17.3(B)

RESULT $F = IlB \sin\theta$

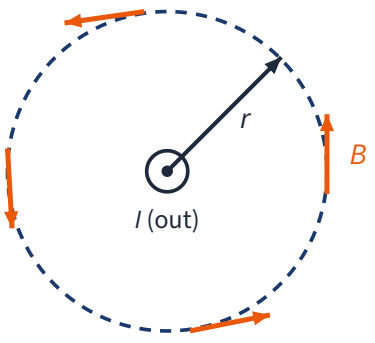
1. Each charge carrier q drifts with velocity v at angle θ to B: force $qvB \sin\theta$.
2. Number of carriers in length l: nAl.
3. Total force $F = (nAl)qvB \sin\theta$.
4. $I = nAqv$, so $\mathbf{F = IlB \sin\theta}$.

3 FIELD OF A LONG STRAIGHT WIRE (AMPÈRE'S LAW)

SYLLABUS 17.4(B)

RESULT $B = \frac{\mu_0 I}{2\pi r}$

1. Ampère's law: $\oint B \cdot dl = \mu_0 I$.
2. Choose a circle of radius r centred on the wire. By symmetry B is tangential and has the same size all round.
3. $\oint B \cdot dl = B(2\pi r)$.
4. $B(2\pi r) = \mu_0 I$, so $\mathbf{B = \frac{\mu_0 I}{2\pi r}}$.



4 FORCE BETWEEN TWO PARALLEL CONDUCTORS

SYLLABUS 17.5(A)

RESULT $F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}$

1. Field of wire 1 at wire 2 (distance d): $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$, perpendicular to wire 2.
2. Force on length l of wire 2: $F = B_1 I_2 l$.
3. $\mathbf{F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}}$; an equal and opposite force acts on wire 1.
4. Currents in the same direction attract; opposite directions repel.

5 E/M IN THOMSON'S EXPERIMENT

SYLLABUS 17.6(C)

RESULT $\dfrac{e}{m} = \dfrac{E}{rB^2}$

1. Crossed E and B fields are adjusted so the beam is undeflected: $eE = evB$, so $v = \dfrac{E}{B}$.
2. With E switched off, the electrons move in a circle of radius r in B: $evB = \dfrac{mv^2}{r}$.
3. $\dfrac{e}{m} = \dfrac{v}{rB}$.
4. Substitute $v = \dfrac{E}{B}$: $\mathbf{\dfrac{e}{m} = \dfrac{E}{rB^2}}$.

Alternative speed selection by energy: $eV = \tfrac{1}{2}mv^2$, so $v = \sqrt{\dfrac{2eV}{m}}$ and $\dfrac{e}{m} = 2V/(B^2r^2)$. State which condition you use.

6 BAINBRIDGE MASS SPECTROMETER

SYLLABUS 17.6

RESULT $m = \dfrac{qBB'r}{E}$

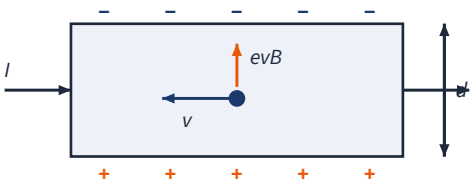
1. **Speed selector**: crossed E and B fields; only ions with $qE = qvB$ pass undeflected, so $v = \dfrac{E}{B}$.
2. **Deflection chamber**: a second field B′ alone; the magnetic force provides the centripetal force: $qvB' = \dfrac{mv^2}{r}$.
3. $r = \dfrac{mv}{qB'} = \dfrac{mE}{qBB'}$, so ions of different masses (isotopes) follow semicircles of different radii.
4. $\mathbf{m = \dfrac{qBB'r}{E}}$, or specific charge $\dfrac{q}{m} = \dfrac{E}{BB'r}$; the separation of two isotopes on the plate is $2\Delta r$.

7 HALL VOLTAGE

SYLLABUS 17.7(B)

RESULT $V_H = \dfrac{BI}{net}$

1. Carriers (charge e, number density n) drift with velocity v through a slab of width d and thickness t, with B perpendicular.
2. The magnetic force evB pushes carriers to one face; charge builds up and sets up a Hall field $E_H = \dfrac{V_H}{d}$.
3. Equilibrium: $\dfrac{eV_H}{d} = evB$, so $V_H = Bvd$.
4. $I = neAv = ne(dt)v$, so $v = I/(nedt)$.
5. $\mathbf{V_H = \dfrac{BI}{net}}$.



B into page; V_H across *d*; thickness *t*

Condition: equilibrium ($F_{electric} = F_{magnetic}$) while the current continues to flow. $V_H = Bvd$ needs the width d; in $V_H = \dfrac{BI}{nqt}$ the Hall voltage depends on the thickness t, not on the width.

CHAPTER 18 · DERIVATIONS

ELECTROMAGNETIC INDUCTION

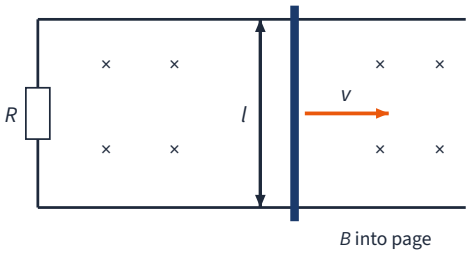
5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 E.M.F. IN A MOVING STRAIGHT CONDUCTOR

SYLLABUS 18.2(C)

RESULT $\varepsilon = Blv \sin\theta$

1. A rod of length l moves with velocity v at angle θ to the field B (rod perpendicular to B).
2. In time dt it sweeps an area $l(v \sin\theta)dt$ across the field lines.
3. Flux cut $d\Phi = Blv \sin\theta\, dt$.
4. Faraday's law: $\boldsymbol{\varepsilon = \frac{d\Phi}{dt} = Blv \sin\theta}$.



Method 2 (charge moving in a magnetic field): each free electron in the rod feels qvB along the rod; charges separate until qE = qvB, so ε = El = Blv.

2 E.M.F. IN A ROTATING COIL

SYLLABUS 18.2(D)

RESULT $\varepsilon = NBA\omega \sin\omega t$

1. A coil of N turns and area A rotates at angular velocity ω about an axis perpendicular to B.
2. Starting with its plane perpendicular to B, the angle between B and the normal is $\theta = \omega t$.
3. Flux linkage $N\Phi = NBA \cos\omega t$.
4. $\boldsymbol{\varepsilon = -\frac{d(N\Phi)}{dt} = NBA\omega \sin\omega t}$; peak value $\varepsilon_0 = NBA\omega$.

3 SELF-INDUCTANCE OF A SOLENOID

SYLLABUS 18.3(D)

RESULT $L = \frac{\mu_0 N^2 A}{l}$

1. Long solenoid: N turns, length l, area A, current I: $B = \frac{\mu_0 NI}{l}$.
2. Flux through each turn $\Phi = BA = \frac{\mu_0 NIA}{l}$.
3. Flux linkage $N\Phi = \frac{\mu_0 N^2 AI}{l}$.
4. $LI = N\Phi$, so $\boldsymbol{L = \frac{\mu_0 N^2 A}{l}}$.

4 BACK E.M.F. IN A COIL

SYLLABUS 18.3

RESULT $\varepsilon = -L\dfrac{dI}{dt}$

1. Flux linkage in the coil is proportional to the current: $N\Phi = LI$.
2. Faraday's law: $\varepsilon = -\dfrac{d(N\Phi)}{dt}$.
3. Differentiate: $\varepsilon = -\dfrac{d(LI)}{dt}$, and L is constant, so $\boldsymbol{\varepsilon = -L\dfrac{dI}{dt}}$.
4. The minus sign (Lenz's law): the back e.m.f. opposes the change in current.

5 MUTUAL INDUCTANCE OF TWO COAXIAL SOLENOIDS

SYLLABUS 18.5(D)

RESULT $M = \dfrac{\mu_0 N_p N_s A}{l_p}$

1. Primary solenoid (N_p turns, length l_p) carries I_p: $B = \dfrac{\mu_0 N_p I_p}{l_p}$.
2. The secondary (N_s turns) is wound coaxially with the same area A: $\Phi_s = BA$.
3. $N_s\Phi_s = \dfrac{\mu_0 N_p N_s A I_p}{l_p}$.
4. $MI_p = N_s\Phi_s$, so $\boldsymbol{M = \dfrac{\mu_0 N_p N_s A}{l_p}}$.

CHAPTER 19 · DERIVATIONS

ALTERNATING CURRENT CIRCUITS

5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 PURE RESISTOR: CURRENT AND POWER

SYLLABUS 19.1(C), (E)

RESULT $I = I_0 \sin \omega t$ (in phase); $\langle P \rangle = \frac{1}{2}V_0 I_0 = I_{rms}^2 R$

1. $V = V_0 \sin \omega t$ across R: $I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = \mathbf{I_0 \sin \omega t}$, in phase with V.
2. Instantaneous power $P = VI = V_0 I_0 \sin^2 \omega t = \frac{1}{2}V_0 I_0 (1 - \cos 2\omega t)$.
3. The mean of $\cos 2\omega t$ over a cycle is zero: $\mathbf{\langle P \rangle = \frac{1}{2}V_0 I_0 = I_{rms} V_{rms} = I_{rms}^2 R}$.

2 PURE INDUCTOR: CURRENT, REACTANCE AND POWER

SYLLABUS 19.2(A), (D), (E)

RESULT $I = I_0 \sin(\omega t - \frac{\pi}{2})$; $X_L = \omega L$; $\langle P \rangle = 0$

1. $V = V_0 \sin \omega t$ is balanced by the back e.m.f.: $V_0 \sin \omega t = L \frac{dI}{dt}$.
2. Integrate: $I = -\frac{V_0}{\omega L} \cos \omega t = \mathbf{I_0 \sin(\omega t - \frac{\pi}{2})}$: the current lags the voltage by $\frac{\pi}{2}$.
3. $X_L = \frac{V_0}{I_0} = \mathbf{\omega L}$.
4. $P = VI = -V_0 I_0 \sin \omega t \cos \omega t = -\frac{1}{2}V_0 I_0 \sin 2\omega t$; its mean over a cycle is **zero**.

3 PURE CAPACITOR: CURRENT, REACTANCE AND POWER

SYLLABUS 19.3(A), (D), (E)

RESULT $I = I_0 \sin(\omega t + \frac{\pi}{2})$; $X_C = \frac{1}{\omega C}$; $\langle P \rangle = 0$

1. $Q = CV = CV_0 \sin \omega t$.
2. $I = \frac{dQ}{dt} = \omega C V_0 \cos \omega t = \mathbf{I_0 \sin(\omega t + \frac{\pi}{2})}$: the current leads the voltage by $\frac{\pi}{2}$.
3. $X_C = \frac{V_0}{I_0} = \mathbf{\frac{1}{\omega C}}$.
4. $P = VI = \frac{1}{2}V_0 I_0 \sin 2\omega t$; its mean over a cycle is **zero**.

4 RESONANCE FREQUENCY OF AN R-L-C CIRCUIT

SYLLABUS 19.4(F)

RESULT $f_0 = 1/(2\pi\sqrt{LC})$

1. $Z = \sqrt{R^2 + (X_L - X_C)^2}$ is smallest ($Z = R$) when $X_L = X_C$.
2. $\omega_0 L = \dfrac{1}{\omega_0 C}$, so $\omega_0 = \dfrac{1}{\sqrt{LC}}$.
3. **$f_0 = 1/(2\pi\sqrt{LC})$**; the current is then maximum and in phase with the supply.

5 IMPEDANCE OF A SERIES R-L-C CIRCUIT (PHASOR METHOD)

SYLLABUS 19.4

RESULT $Z = \sqrt{R^2 + (X_L - X_C)^2}$

1. The same current I flows through R, L and C. Draw the current phasor along the reference axis.
2. $V_R = IR$ is in phase with I; $V_L = IX_L$ leads I by 90°; $V_C = IX_C$ lags I by 90°.
3. Add the phasors: $V^2 = V_R^2 + (V_L - V_C)^2 = I^2[R^2 + (X_L - X_C)^2]$.
4. **$Z = \dfrac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$**, and the phase angle is $\tan\varphi = \dfrac{X_L - X_C}{R}$.
5. For R-C only: $Z = \sqrt{R^2 + X_C^2}$; for R-L only: $Z = \sqrt{R^2 + X_L^2}$.